

An analytic expression for the composition of aqueous HNO_3 - H_2SO_4 stratospheric aerosols including gas phase removal of HNO_3

Kenneth S. Carslaw, Beiping Luo and Thomas Peter

Max-Planck-Institut für Chemie, Mainz, Germany.

Abstract. An analytic expression is given for the composition and volume of stratospheric HNO_3 - H_2SO_4 - H_2O aerosols in terms of the temperature and total amounts of H_2O , HNO_3 and H_2SO_4 , taking into account conservation of HNO_3 . In contrast to previous parameterisations of equilibrium vapour pressures, the present scheme for calculating the aerosol composition does not require an iteration procedure, thus computation times are significantly reduced, making it ideal for inclusion in large-scale atmospheric models. Calculated compositions are compared with the results of a comprehensive thermodynamic model for all stratospheric conditions and maximum deviations are presented.

Introduction

Thermodynamic model calculations show that stratospheric aqueous H_2SO_4 aerosols absorb large amounts of HNO_3 and H_2O from the gas phase for temperatures between the nitric acid trihydrate (NAT) and ice frost points (Carslaw *et al.*, 1994; Tabazadeh *et al.*, 1994), leading to a steep increase in volume and a change in the composition from almost pure H_2SO_4 - H_2O to an aerosol composed principally of HNO_3 and H_2O . Due to this uptake, HNO_3 can be strongly depleted from the gas phase, particularly during volcanically perturbed periods with increased aerosol amounts. The calculated change in volume was shown by Carslaw *et al.* (1994) and Drdla *et al.* (1994) to agree closely with *in-situ* observations made in the Arctic during the winter of 1989 (Dye *et al.*, 1992), leading to the conclusion that aqueous HNO_3 - H_2SO_4 aerosols may remain liquid for temperatures as low as the ice frost point. This is supported by laboratory experiments (Koop *et al.*, 1995) which show that even bulk supercooled solutions are unlikely to freeze under such conditions.

The recognition that stratospheric aerosols may remain liquid at polar winter temperatures leads to a need for reliable methods of estimating the aerosol composition, both for the interpretation of particle observations and for use within stratospheric chemical models. We have therefore derived an analytic expression for the composition of ternary aerosols and the gaseous HNO_3 concentration in terms of the temperature and total amounts of H_2SO_4 , HNO_3 and H_2O , based upon the thermodynamic model of Carslaw *et al.* (1995).

Analytic Expression for the Aerosol Composition

The solubility of a gas phase species in an aerosol can be calculated from its effective Henry's law constant, $H_i^* = c_i/p_i$ where c_i is the concentration of component i in the aerosol,

p_i is the partial vapour pressure and H_i^* is a function of the aerosol composition and temperature. For species of low abundance, for example HCl or HBr, even complete dissolution leads to only a minor change in the aerosol composition. The solubility of such a species is therefore not affected by its presence in the aerosol, and to a good approximation the partitioning can be calculated simply in terms of H_i^* , combined with the principle of mass conservation. However, the uptake of HNO_3 leads to a marked change in the aerosol composition (Carslaw *et al.*, 1994). Because of the complex dependence of solubility on concentrations (Carslaw *et al.*, 1995; Luo *et al.*, 1995; Tabazadeh *et al.*, 1994), the partitioning must be calculated using an iteration procedure, which is time consuming within a large atmospheric model. We have previously argued that aerosol composition should be calculated using self-consistent thermodynamic models, based upon a wide range of thermodynamic data (Carslaw *et al.*, 1995; Luo *et al.*, 1995). Such model expressions allow reliable extrapolations (for example of the vapour pressures) outside the range of available data, but are too complex to be solved analytically to obtain the aerosol composition. The simpler equations presented here can be solved without the need for iteration and were adopted based upon the agreement with our self-consistent model expressions (Carslaw *et al.*, 1995), which are in excellent agreement with available thermodynamic data for the HNO_3 - H_2SO_4 system, including vapour pressures of HNO_3 and H_2O to less than 200 K.

The composition expression is valid over the range of water partial pressure (p_w) and temperature

$$2 \times 10^{-5} \text{ mb} < p_w < 2 \times 10^{-3} \text{ mb} \\ \max\{T_{\text{ice}} - 3\text{K}, 185\text{K}\} < T < 240 \text{ K}, \quad (1)$$

where T_{ice} is the ice frost point. It is assumed that other soluble species, for example HCl, are trace components of the HNO_3 - H_2SO_4 - H_2O aerosol and therefore do not affect the uptake of HNO_3 , which is a reasonable assumption for stratospheric conditions. Secondly, we assume that the water partial pressure remains constant during the uptake of HNO_3 , since the amount of water in the aerosol is negligible compared to the total amount.

The basic relation of interest is the conservation of mass of HNO_3 between the gas and aerosol phases

$$(p_N^0 - p_N)/RT = n_S m_N/m_S, \quad (2)$$

where p_N (atm) is the equilibrium vapour pressure of HNO_3 over the droplet and p_N^0 (atm) is the total partial pressure of HNO_3 assuming no removal from the gas phase. The difference between these is the amount of HNO_3 in the aerosol, and this is expressed by the term on the right hand side in terms of the molality ($\text{mol kg}^{-1} \text{ H}_2\text{O}$) of HNO_3 (m_N) and H_2SO_4 (m_S) and the total amount of H_2SO_4 per unit

Copyright 1995 by the American Geophysical Union.

Paper number 95GL01668

0094-8534/95/95GL-01668\$03.00

volume of air (n_s , mol m⁻³). The gas constant, $R = 8.207 \times 10^{-5}$ m³ atm mol⁻¹ K⁻¹ and $T(K)$ is the temperature. The equilibrium vapour pressure of HNO₃ is given by

$$p_N = m_N/H^*, \quad (3)$$

where H^* is an effective Henry's law constant (mol kg⁻¹ atm⁻¹) describing the equilibrium between total dissolved HNO₃ (NO₃⁻ + HNO₃) and the vapour. There are therefore four unknown quantities to be determined - p_N , m_N , m_s and H^* . The quantity H^* is unknown since this depends upon the composition of the aerosol, and therefore on m_N and m_s . Hence two further equations are required to obtain a solution. The first relates the Henry's law constant in the ternary solution to that in the two binary (HNO₃-H₂O and H₂SO₄-H₂O) solutions at the same temperature and water vapour pressure

$$H^* = H_{NNb}^* m_N / (m_N + m_s) + H_{NSb}^* m_s / (m_N + m_s). \quad (4)$$

In this equation H_{NNb}^* is the Henry's law constant of HNO₃ in a binary HNO₃-H₂O solution at the water vapour pressure and temperature of interest and H_{NSb}^* is the Henry's law constant of HNO₃ in a binary H₂SO₄-H₂O solution (with HNO₃ a trace species). This empirical expression for the solubility of HNO₃, in terms of a molality-weighted linear combination of Henry's law constants, is in good agreement with the model expressions of Carslaw *et al.* (1995) for stratospheric conditions.

The final equation relates the concentrations of HNO₃ and H₂SO₄ in the ternary solution to those in the two binary solutions (m_{Nb} and m_{Sb}) for the same water vapour pressure and temperature (Stokes and Robinson, 1966)

$$m_s / m_{Sb} + m_N / m_{Nb} = 1. \quad (5)$$

Note that m_{Sb} is simply the concentration of a binary H₂SO₄-H₂O aerosol in the stratosphere for a given water partial pressure and temperature. One assumption of Eqn. (5) is that

the interaction between the solute species is insignificant compared to the interaction between each of the solutes and the solvent (*i.e.*, those interactions arising in the two binary solutions) (Stokes and Robinson, 1966). It has previously been shown (Carslaw *et al.*, 1995) that solute-solute interactions in the HNO₃-H₂SO₄-H₂O system may be important in concentrated H₂SO₄ solutions (greater than 50 wt% H₂SO₄). However, the solubility of HNO₃ in these solutions is generally low and for stratospheric conditions we expect the assumptions inherent in Eq. (5) to have little effect on the calculated composition. This is shown to be the case. Test calculations showed that the binary solution parameters H_{NNb}^* , H_{NSb}^* , m_{Nb} and m_{Sb} in Eqs. (4-5) could be fitted as simple expressions in terms of temperature and water vapour pressure alone. These are given in the next section.

The solution of Eqs. (2-5) yields a third order equation in terms of m_s or m_N . The analytic solution is

$$m_s = - (A + 2\cos([\pi + \phi]/3) (A^2 - 3B)^{1/2})/3 \quad (6)$$

and from Eq. (5), the HNO₃ concentration in the liquid is

$$m_N = m_{Nb} (1 - m_s / m_{Sb}), \quad (7)$$

while the gas phase HNO₃ concentration (p_N) can be calculated from Eqs. (3) and (4). The parameters ϕ , A and B are functions of the quantities T , n_s , p_N^0 , H_{NNb}^* , H_{NSb}^* , m_{Nb} and m_{Sb} . Because these functions are lengthy, we give them in the form of general computer code in Table 1 for straightforward inclusion in numerical models.

Binary Solution Parameters

The effective Henry's law constant of HNO₃ in the binary solutions HNO₃-H₂O and H₂SO₄-H₂O were fitted to the model calculations of Carslaw *et al.* (1995) according to the expression

Table 1. Code for calculating the composition of aqueous HNO₃-H₂SO₄ aerosols

```
;Inputs:T(K), NS(total H2SO4, mol m-3), PNO (total HNO3, atm)
;Inputs:MNB, MSB (binary solution concentrations /mol (kg H2O)-1) (Eq. (10) in text)
;Inputs:HNNB,HNSB (binary solution effective Henry constants /mol (kg H2O)-1 atm-1 (Eq. (8))
;Constants: R (8.205e-5 m3 atm mol-1 K-1)
TT = R*T*NS
A = (TT*HNNB*MNB**2-TT*HNSB*MNB*MSB-2*MNB**2*MSB*MNB*MSB**2 +
HNNB*MNB*MSB*PNO-HNSB*MSB**2*PNO) / (MNB**2-MNB*MSB)
B = MSB*(-2*TT*HNNB*MNB+TT*HNSB*MSB+MNB*MSB-HNNB*MSB*PNO) / (MNB-MSB)
C = (TT*HNNB*MNB*MSB**2) / (MNB - MSB)
PHI = ATAN(SQRT(4*(A**2-3*B)**3-(-2*A**3+9*A*B-27*C)**2)/(-2*A**3+9*A*B-27*C))
PI = 2*ASIN(1.0)
IF (PHI.LT.0) PHI=PHI+PI
MS = -1/3*(A+2*SQRT(A**2-3*B)*COS((PI+PHI)/3))
MN = MNB*(1-MS/MSB)
PN = MN/(HNNB*MN/(MN+MS)+HNSB*MS/(MN+MS))
WS = MS*0.09812/(1+MS*0.09812+MN*0.06303)
WN = MN*0.06303/(1+MS*0.09812+MN*0.06303)
RHOS = 1000+(123.64-5.6e-4*T**2)*MS-(29.54-1.81E-4*T**2)*MS**1.5
+(2.343-1.487E-3*T-1.324E-5*T**2)*MS**2
RHON = 1000+(85.11-5.04e-4*T**2)*MN-(18.96-1.427e-4*T**2)*MN**1.5
+(1.458-1.198e-3*T-9.703e-6*T**2)*MN**2
RHO = 1/((1/RHOS)*MS/(MN+MS)+(1/RHON)*MN/(MN+MS))
V = NS*0.09812/(WS*RHO)
```

For program verification: With 5 ppmv H₂O, 10 ppbv HNO₃, 0.5 ppbv H₂SO₄ at $T = 192$ K and 50 mb pressure, $WS=0.262$ and $WN=0.226$. Note that for $T > 215$ K or $PNO = 0$, $MS = MSB$ and $MN = 0$ and this program is not required to calculate MN and MS .

Table 2. Coefficients for the calculation of binary solution parameters

Coefficient	q_1, k_1	q_2, k_2	q_3, k_3	q_4, k_4	q_5, k_5	q_6, k_6	q_7, k_7	q_8	q_9	q_{10}
H_{NNb}^*	14.5734	0.0615994	-1.14895	0.691693	-0.098863	0.0051579	0.123472	-0.115574	0.0110113	0.0097914
H_{NSb}^*	14.4700	0.0638795	-3.29597	1.778224	-0.223244	0.0086486	0.536695	-0.335164	0.0265153	0.0157550
m_{Nb}	-3.9136(1)	6.3584(3)	8.329(1)	-1.7650(4)	1.9853(2)	-1.1948(4)	-2.8469(1)	-	-	-
m_{Sb}	-2.1661(1)	2.7242(3)	5.181(1)	-1.5732(4)	4.7004(1)	-6.9690(3)	-4.6183(0)	-	-	-

Coefficients for H_{NNb}^* and H_{NSb}^* (Eq. (8) in the text) are valid for $\max\{T_{\text{ice}} - 3\text{K}, 185\text{K}\} < T < 215\text{K}$, where T_{ice} is the ice frost point, and for water vapour pressures given in Eq. (1). Coefficients for the calculation of $m_{\text{Nb}}/\text{mol kg}^{-1}$ and m_{Sb} (Eq. (10) in the text) are valid over the range of temperature: for m_{Sb} : $\max\{T_{\text{ice}} - 3\text{K}, 185\text{K}\} < T < 240\text{K}$ and for m_{Nb} : $\max\{T_{\text{ice}} - 3\text{K}, 185\text{K}\} < T < 215\text{K}$. For 6.3584(3) read 6.3584×10^3 .

$$\ln(H_{\text{NNb}}^*, H_{\text{NSb}}^*/\text{mol kg}^{-1} \text{ atm}^{-1}) = q_1 + q_2 T_r^2 + (q_3 + q_4 T_r + q_5 T_r^2 + q_6 T_r^3) \cdot p_r + (q_7 + q_8 T_r + q_9 T_r^2) \cdot p_r^2 + q_{10} T_r p_r^3, \quad (8)$$

where $T_r = 10^4(1/T - 1/230)$ and $p_r = \ln(p_w) + 18.4$. The parameters q_{1-10} are given in Table 2.

The binary solution molalities can be obtained from simplified expressions for the water vapour pressure, which is related to the vapour pressure over pure (supercooled) water (p_w^0) and the water activity (a_w) by $p_w = p_w^0 \times a_w$. Noting the temperature variation of $\ln(p_w^0)$ and by representing $\ln(a_w)$ as a second order polynomial in concentration, we have fitted water vapour pressures from the model of Carslaw *et al.* (1995) to the expression

$$\ln(p_w/\text{atm}) = (k_1 + k_2/T) \cdot x + (k_3 + k_4/T) \cdot x^2 + k_5 + k_6/T + k_7 \cdot \ln(T), \quad (9)$$

where x is the mole fraction of HNO₃ or H₂SO₄ in a binary solution and the concentration is given by $m = 55.51 (\text{mol kg}^{-1}) x/(1-x)$. Equation (9) yields the following expression for the concentration of a binary solution

$$x_{\text{Nb}}, x_{\text{Sb}} = \{-k_1 - k_2/T - [(k_1 + k_2/T)^2 - 4(k_3 + k_4/T) \times (k_5 + k_6/T + k_7 \ln(T) - \ln(p_w))]^{1/2}\} / 2(k_3 + k_4/T). \quad (10)$$

The coefficients k_{1-7} are listed in Table 2. Because our principal aim is to develop a reliable expression for the composition of HNO₃-H₂SO₄-H₂O aerosols, the parameters related to the solubility of HNO₃ (m_{Nb} , H_{NNb}^* and H_{NSb}^*) were fitted only for temperatures less than 215 K. This is well above the temperature at which significant uptake of HNO₃ occurs, even for the highest possible stratospheric partial pressures of HNO₃ and H₂O. The expression for m_{Sb} was fitted to 240 K so that the equations successfully predict the composition of binary H₂SO₄-H₂O aerosols upto this temperature, with HNO₃ present only as a trace species.

Stratospheric Calculations

The gas and aqueous phase concentrations of any other soluble trace species, HX, for example HCl, HBr, HOCl or HOBr, can now be calculated using an appropriate expression for the effective Henry's law constant (in terms of aqueous HNO₃ and H₂SO₄ concentrations) and an expression for mass conservation similar to Eq. (2), yielding

$$m_{\text{HX}} = p_{\text{HX}}^0 / (RTn_s/m_s + 1/H_{\text{HX}}^*), \quad (11)$$

where p_{HX}^0 (atm) is the total partial pressure of HX (assuming no gas phase removal) and H_{HX}^* (mol kg⁻¹ atm⁻¹) is the effective Henry's law constant in the ternary HNO₃-H₂SO₄-H₂O solution. At present there are no measurements of the solubility of important trace species in cold, highly concentrated HNO₃-H₂SO₄ aerosols. However, the Henry's law constants of HCl and HBr can be obtained from the thermodynamic model expressions for the vapour pressures given by Luo *et al.* (1995) in terms of HNO₃ and H₂SO₄ concentrations, and for HOCl we recommend the expression from Huthwelker *et al.* (1995): $\ln(H_{\text{HOCl}}^*/\text{mol kg}^{-1} \text{ atm}^{-1}) = 6.4946 + (m_{\text{N}} + m_{\text{S}}) \times (0.04107 - 54.56/T) - 5862(1/298.15 - 1/T)$. The gas phase abundance (p_{HX}) is lower than p_{HX}^0 due to partitioning to the aerosol and can be calculated from Henry's law in the form of Eq. (3).

Another quantity of interest is the volume of the aerosol per unit volume of air, given by

$$v = n_s M_s / w_s \rho, \quad (12)$$

where w_s is the weight fraction of H₂SO₄ in the aerosol ($w_s = m_s M_s / (1 + M_s m_s + M_N m_N)$) and M_s and M_N are the molar masses of H₂SO₄ and HNO₃ (kg mol⁻¹). Ternary solution mass densities, ρ , can be estimated in terms of a molality weighted combination of binary solution densities, ρ_s and ρ_N ,

$$1/\rho = (1/\rho_s) m_s / (m_s + m_N) + (1/\rho_N) m_N / (m_s + m_N). \quad (13)$$

We have fitted ρ_s and ρ_N as simple functions of concentration and temperature using tabulated densities (Sohnel and Novotny, 1985) for concentrations between 5 and 80 wt%. These are given in Table 1.

Finally, we note that it is common in some laboratory investigations to employ constant partial pressures of both H₂O and HNO₃. The calculation for fixed p_N can be achieved by setting $n_s = 0$. Note, however, that below the HNO₃-H₂O dew point equilibrium cannot be established for a fixed HNO₃ partial pressure (physically, such a solution would grow indefinitely by condensation).

Comparison with Thermodynamic Model

There are a number of assumptions in the derivation of Eqs. (6) and (7). We have therefore compared predicted compositions with those calculated using the comprehensive model expressions of Carslaw *et al.* (1995) for a very wide range of stratospheric conditions, and calculate the following maximum deviations

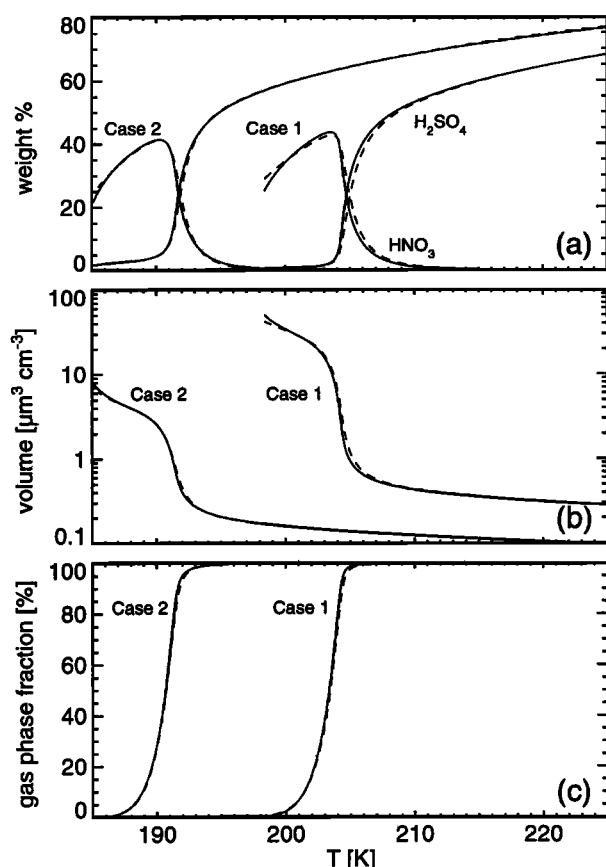


Figure 1. Calculated composition (a), volume (b) and gas phase HNO_3 fraction (c) of aqueous HNO_3 - H_2SO_4 aerosols. Solid line - thermodynamic model of Carslaw *et al.* (1995); dashed line - analytic equations from this work. Case 1: $p=200$ mb, $\text{HNO}_3 = 20$ ppbv, $\text{H}_2\text{O} = 10$ ppmv and $\text{H}_2\text{SO}_4=0.3$ ppbv (conditions chosen to demonstrate the largest deviations between model and analytic expression). Case 2: $p = 50$ mb, $\text{HNO}_3 = 10$ ppbv, $\text{H}_2\text{O} = 5$ ppmv and $\text{H}_2\text{SO}_4 = 0.5$ ppbv.

$$\begin{array}{l}
 0.1 < \text{ppbv } \text{H}_2\text{SO}_4 < 100 \\
 0 < \text{ppbv } \text{HNO}_3 < 20 \\
 1 < \text{ppmv } \text{H}_2\text{O} < 10 \\
 20 < p(\text{mb}) < 200 \\
 \max\{T_{\text{ice}}-3, 185\} < T(\text{K}) < 240
 \end{array}
 \Rightarrow
 \begin{array}{l}
 (\Delta \text{HNO}_3)_{\text{max}} = 7 \text{ wt\%} \\
 (\Delta \text{H}_2\text{SO}_4)_{\text{max}} = 7 \text{ wt\%} \\
 (\Delta V/V)_{\text{max}} = 80\% \\
 (\Delta \text{HNO}_3^{\text{gas}}) = 10\%
 \end{array}$$

The largest deviation from the thermodynamic model occurs at the limit of the range of conditions given above - at 200 mb with 20 ppbv HNO_3 and 10 ppmv H_2O . This is shown in Fig. (1) (Case 1) for 0.3 ppbv H_2SO_4 , but deviations are similar for a wide range of H_2SO_4 amounts (Note, however, that the deviation is much reduced for very high H_2SO_4 amounts because the aerosol is then composed principally of H_2SO_4 and water for all temperatures, for

which the model and analytic expression are in excellent agreement). The maximum deviation from the model always occurs for conditions where the concentration of both HNO_3 and H_2SO_4 in the aerosol are high. It is clear that, although the deviation reaches about 7 wt% for both species, the difference is small when considered in terms of the temperature change required to alter the composition by this amount; in this case, a change of only 0.5 K is sufficient to account for the deviation between the model and the analytic expression, which is unlikely to be significant within numerical atmospheric models, where temperature fields are not known with this precision. These conditions are extreme for the stratosphere. For more realistic conditions (Case 2) the maximum deviation is about 4 wt%, equivalent to a change in temperature of only 0.2 - 0.3 K.

Notes & Acknowledgments. Part of this work was supported by the German BMBF under 01 LO 9221/4. A FORTRAN computer code for these calculations is available from the authors.

References

- Carslaw, K.S., *et al.*, Stratospheric aerosol growth and HNO_3 gas phase depletion from coupled HNO_3 and water uptake by liquid particles, *Geophys. Res. Lett.*, **21**, 2479-2482, 1994.
- Carslaw, K.S., *et al.*, A thermodynamic model of the system HCl - HNO_3 - H_2SO_4 - H_2O , including solubilities of HBr , from <200 K to 328 K, *J. Phys. Chem.*, in press, 1994.
- Drdla, K., *et al.*, Analysis of the physical state of one Arctic polar stratospheric cloud based on observations, *Geophys. Res. Lett.*, **21**, 2475-2478, 1994.
- Dye, J.E., *et al.*, Particle size distributions in Arctic polar stratospheric clouds, growth and freezing of sulfuric acid droplets, and implications for cloud formation, *J. Geophys. Res.*, **97**, 8015-8034, 1992.
- Huthwelker, T., *et al.*, Solubility of HOCl in water and aqueous H_2SO_4 to stratospheric temperatures, *J. Atmos. Chem.*, in press, 1995.
- Koop, T., *et al.*, Do stratospheric aerosol droplets freeze above the ice frost point?, *Geophys. Res. Lett.*, **22**, 917-920, 1995.
- Luo, B.P., *et al.*, Vapour pressures of H_2SO_4 / HNO_3 / HCl / HBr / H_2O solutions to low stratospheric temperatures, *Geophys. Res. Lett.*, **22**, 247-250, 1995.
- Sohnel, O. and P. Novotny, *Densities of Aqueous Solutions of Inorganic Substances*, Elsevier, Amsterdam, 1985.
- Stokes, R.H. and R.A. Robinson, Interactions in aqueous nonelectrolyte solutions. I. Solute-solvent equilibria, *J. Phys. Chem.*, **70**, 2126-2130, 1966.
- Tabazadeh, A., *et al.*, A model for studying the composition and chemical effects of stratospheric aerosols, *J. Geophys. Res.*, **99**, 12,897-12,914, 1994.

K.S. Carslaw, B. Luo and Th. Peter, Max-Planck Institut für Chemie, P.O. Box 3060, D-55020 Mainz, Germany.
email carslaw@nike.mpch-mainz.mpg.de

(Received March 13, 1995; revised May 15, 1995;
accepted June 2, 1995)